

Elbow Method

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```
1 install.packages("dplyr")
2 install.packages("ggplot2")
3 install.packages("ggfortify")
4 library("ggplot2")
5 library("dplyr")
6 library("ggfortify")
7
8 data <- select(iris, c(1:4))
9 set.seed(123)
10
11 k.max <- 15
12 wss <- sapply(1:k.max,
13               function(k){kmeans(data, k, nstart=50, iter.max = 15 )$tot.withinss})
14 wss
15 plot(1:k.max, wss,
16       type="b", pch = 19, frame = FALSE,
17       xlab="Number of clusters K",
18       ylab="Total within-clusters sum of squares")
19
20
```